

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Assessment Tool 1 – Application based questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	A14ML 3rd year	Date:	17/07/2026

Instructions to Students

- Identify the relevant computer architecture concepts, functional units, or performance parameters required to solve the given application-based problem.
- Analyse the interaction between CPU, memory, bus structures, and I/O components in the given scenario.
- Apply appropriate instruction execution cycles, addressing modes, and architectural concepts to determine the correct solution.
- Compute performance metrics such as CPI, clock cycles, CPU execution time, speedup, or benchmarking measures using appropriate formulas and calculations.
- Interpret the calculated results and explain their impact on overall computer system performance.

QUESTIONS

1. A programmer wants to understand how data is transferred between registers during program execution in an 8085 microprocessor. To observe register operations during program execution, write and execute an 8085 program that performs data transfer and arithmetic operations using multiple registers. Observe the contents of the Accumulator and the general-purpose registers (B, C, D, E, H, and L) after each instruction. Analyse how register values change during execution and determine the role of each register in completing the program.
2. A user notices that a computer responds slowly while attending an online discussion, editing a document, and downloading files simultaneously. To analyse the system behaviour under multitasking, observe the Memory usage, Disk activity and Network activity while performing these tasks. Analyse how the CPU, memory, and I/O devices cooperate during multitasking, and determine which hardware resource is most likely to become the performance bottleneck.
3. A software engineer wants to identify the most suitable computer for running engineering applications. To compare the hardware capabilities, observe the Processor Generation, Base Speed, Current Speed, Virtualization Status, Installed RAM, Cache Memory (L1/L2/L3 if available), and Storage Type (SSD/HDD) for two computers. Analyse the observed specifications to determine which system is expected to execute programs more efficiently, and justify your conclusion by relating these hardware characteristics to processor performance.

Code of Conduct

I Kolar Umme Sadiya (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sadiya

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Q. No	Understanding of Problem Context (20)	Application of Concepts (30)	Problem-Solving Approach (20)	Accuracy of Solution (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 30	/ 20	/ 20	/ 10	/ 100

Course Coordinator

Faculty Incharge

Question 1

Study of Register Transfer in 8085 Microprocessor

Aim

To observe how data is transferred between registers during the execution of an 8085 program and analyze the changes in register contents after each instruction.

Program

MVI B,25H

MVI C,35H

MOV A,B

ADD C

MOV D,A

HLT

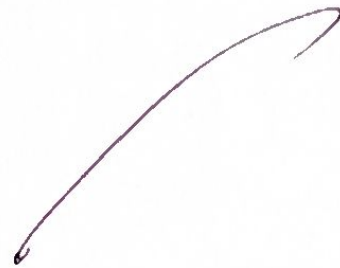
Explanation of Instructions

1. MVI B,25H

Loads the hexadecimal value 25H into register B.

Register Values

Register	Value
A	00H
B	25H
C	00H
D	00H
E	00H
H	00H



Register	Value
L	00H

Observation

Register B changes from 00H to 25H.

2. MVI C,35H

Loads **35H** into register C.

Register	Value
A	00H
B	25H
C	35H
D	00H
E	00H
H	00H
L	00H

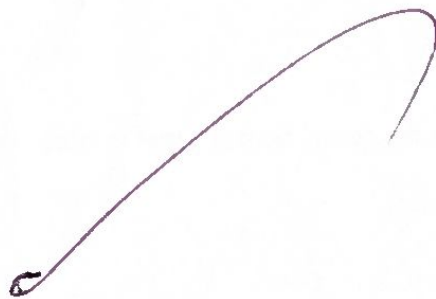
Observation

Register C stores the value 35H.

3. MOV A,B

Copies data from register B into Accumulator.

Register	Value
A	25H
B	25H
C	35H



Observation

Accumulator now contains 25H.

4. ADD C

Adds register C to Accumulator.

$$25H + 35H = 5AH$$

Register	Value
A	5AH
B	25H
C	35H

Observation

Accumulator stores the addition result.

5. MOV D,A

Copies Accumulator to register D.

Register	Value
D	5AH

Observation

Register D receives the final result.

6. HLT

Stops program execution.

Result

The execution shows how data is transferred between registers and how arithmetic operations update register values.

Question 2

CPU Analysis using Resource Monitor

Aim

To analyze CPU, Memory, Disk, and Network usage while multiple applications are running.

CPU Screenshot

(Paste Resource Monitor CPU Screenshot)

Observation

- CPU Usage : 16%
- Maximum Frequency : 85%
- SearchApp.exe is running.
- msedgewebview2.exe processes are visible.
- CPU is executing multiple tasks simultaneously.

Explanation

The CPU performs calculations and executes instructions. During multitasking, the operating system distributes processor time among different applications.

Low CPU utilization indicates the processor is not heavily loaded.

Memory Analysis

(Paste Memory Screenshot)

Observation

- Installed RAM: 8 GB
- Used Memory: 5.1 GB
- Available Memory: 2.8 GB

Explanation

RAM temporarily stores programs and data currently in use.

With 8 GB memory, multitasking applications and several other applications can perform in memory simultaneously.

Disk Analysis

(Paste Disk Screenshot)

Observation

- Disk Read Activity
- Disk Write Activity
- Response Time
- Active Time

Explanation

The storage device permanently stores files and applications.

SSD provides faster read/write speed than HDD, resulting in quicker application loading.

Network Analysis

(Paste Network Screenshot)

Observation

- Data Sent
- Data Received
- Network Utilization
- Active Connections

Explanation

The network adapter exchanges data between the computer and the internet.

Higher utilization occurs while downloading, streaming, or browsing.

Performance Bottleneck

A bottleneck occurs when one hardware component limits the overall system performance.

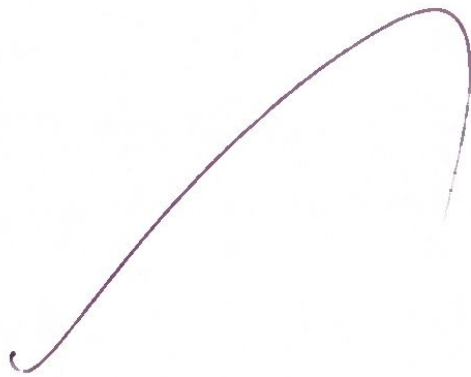
Possible bottlenecks include:

- High CPU Usage
- Low Available RAM
- Heavy Disk Activity
- Slow Network Speed

From the observed screenshots:

- CPU usage is low.
- Memory is sufficient.
- Disk activity is normal.
- Network activity depends on internet usage.

Therefore, no major bottleneck is observed.



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Question 3

Comparison of Two Computer Systems

Specification	System 1	System 2
Processor	Intel Core i5-8250U	Intel Core i7-1165G7
Generation	8th Gen	11th Gen
Base Speed	1.60 GHz	2.80 GHz
Current Speed	1.69 GHz	2.90 GHz
Installed RAM	8 GB	16 GB
L1 Cache	256 KB	512 KB
L2 Cache	1 MB	1.25 MB
L3 Cache	6 MB	12 MB
Virtualization	Enabled	Enabled
Storage	256 GB SSD	512 GB SSD

Comparison

Processor

System 2 has a newer processor generation and better architecture.

RAM

System 2 has double the RAM, making multitasking smoother.

Cache Memory

Larger cache memory reduces processor access time and improves execution speed.

Storage

Both systems use SSDs, but System 2 has higher storage capacity.

Virtualization

Virtualization is enabled in both systems, allowing virtual machines to run efficiently.

Conclusion

System 2 provides better overall performance because it has:

- Newer processor generation
- Higher clock speed
- Larger cache memory
- More RAM
- Larger SSD storage

Therefore, System 2 is more suitable for programming, virtualization, gaming, and multitasking.

Final Conclusion

This assessment helped in understanding:

- Register transfer operations in the 8085 microprocessor.
 - CPU resource utilization using Resource Monitor.
 - Memory, Disk, and Network performance analysis.
 - Performance comparison between two computer systems based on processor, RAM, cache memory, virtualization, and storage
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